

Question Paper

Exam Date & Time: 20-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: CSS 2101/CSS_2101 DATA STRUCTURES

(SPL: COMPUTER SCIENCE AND ENGINEERING / ARTIFICIAL INTELLIGENCE / CYBER SECURITY / DATA SCIENCE / INFORMATION TECHNOLOGY)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) An AVL Tree is created by inserting the following keys in Chronological order: (5)
O, G, T, E, L, Y, Z.
After successful insertion, the key G is deleted.
Analyze and show:
i. The structure of the AVL tree before and after deletion.
ii. Identify the type of imbalance. Specify the rotation(s) used to restore balance.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 1B) Suppose two queues, Q1 and Q2, are implemented in a single array of size 10 using the method (3)
where Q1 grows from left to right and Q2 grows from right to left. Initially, both are empty. If 4
elements are enqueued into Q1 and 3 elements into Q2, what are the possible indices of front1 and
front2? Demonstrate your answer with a neat diagram.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 1C) Write a C program to define a structure Student containing another structure Date (for date of birth). (2)
Read and display the student details using nested structures. The structure should resemble the
following table:

ID	Student Name	DoB		
		DD	MM	YYYY
1	Ram	01	01	2020
2	Lakshman	01	01	2022
3	Hanuma	01	01	2025

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2A) A college is developing a simple "Smart Editor" software for its students. (5)
In the **Smart Editor**, students should be able to **undo** and **redo** text changes. As a software
developer, you are asked to analyze and illustrate how **Stacks** can be used in this application.
Question:
i. Explain how a **Stack** can be used to implement the *Undo-Redo* feature in the editor. Write short C
code snippets for implementation and explain how it will work.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2B) Consider the following example to create binary search tree: (3)
40, 20, 10, 30, 60, 50, 70.
Illustrate with the above BST, how a null pointer is replaced to convert it into a double threaded binary tree.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 2C) Assume that you are storing marks of students in 3 subjects. (2)
Write a C program that:
i. Takes the number of students as input.
ii. Dynamically allocates a **2D array** using pointers.
iii. Accepts marks for each subject. Calculates and prints the **average marks** of each student using only **pointer arithmetic**.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 3A) i. Write a function to perform the Fast Transpose computation of a given sparse matrix. (5)
ii. Demonstrate complete step-by-step process to compute the transpose of the below given sparse matrix given in Fig.Q.3A using the above function. Also show the contents of all the variables (**row_terms** and **starting_pos**) used during the computation process.

0	0	3	0	4	0
0	0	5	7	0	0
0	0	0	0	0	9
0	2	6	0	0	0
1	0	0	0	0	8
0	0	4	0	0	0

Fig.Q.3A

Course Outcome: "CO1" / Cognitive Level: "Application"

- 3B) In a gaming leaderboard system, players' scores are organized in a Binary Search Tree (BST) to (3)
enable quick ranking and retrieval. Write an iterative function to insert a new player's score into the BST.
Insert the score 65 into the following BST and illustrate the updated tree structure after insertion.

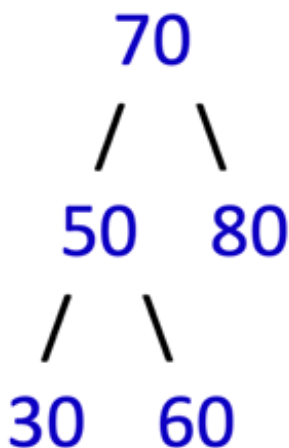


Fig. Q. 3B

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3C) With the help of the following code, justify why a stack is preferred over a queue in handling nested (2)
function calls during program execution. Also, explain the role of activation records (stack frames) in this process
`void fun1() { printf("Inside fun1n"); }`

```
void fun2() { fun1(); printf("Inside fun2n"); }
int main() { fun2(); printf("Inside mainn"); }
```

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4A) A music streaming service keeps track of songs in a playlist using a singly linked list. Each node stores the following details: SongID (integer), Title (string), Duration (in seconds). (5)
- Write a C program to perform the following operations:
 Insert songs into the playlist in the order they are added.
 Remove all songs whose duration is less than a user-specified value (Use head pointer for performing this operation).
 Display the updated playlist.
 Include a main() function to demonstrate all functionalities.

Note:- Use the following prototypes:-

```
typedef struct node {
    int songID;
    char title[50];
    int duration;
    struct node *next;
} Node;
```

```
Node* insertSong(Node *head, int songID, char title[], int duration);
Node* removeShortSongs(Node *head, int minDuration);
void displayPlaylist(Node *head);
```

Course Outcome: "CO2" / Cognitive Level: "Application"

- 4B) Build the expression tree for the given infix expression $8+4*2/(3+1)-5$. Clearly show all the intermediate steps (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) You are provided with the array representation of a binary tree. (2)
- Array Representation: [1, 4, 2, 10, 5, 9, 13] Array index starts with 1.
- Using the given representation, apply your understanding of tree structures to answer the following:
- Construct the binary tree from the given array and determine the depth of node 9.
 - State the formula to find the index of right child given the index of parent and use it to find the index of right child of node 2.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5A) A small delivery network is represented as an undirected graph. The network has six hubs labelled A, B, C, D, E, F. The direct connections between hubs are: A-B, A-C, B-D, C-E, D-F, E-F, B-C, D-E (5)
- Note: Whenever there is a choice between multiple unvisited neighbours in DFS or BFS, hubs should be selected in alphabetical order.
- If a delivery inspection follows Depth First Search (DFS) starting from hub A, which hub will be inspected last? Explain your reasoning.
 - If a delivery inspection follows Breadth First Search (BFS) starting from hub A, which hub is the farthest from A in terms of the number of connections?
 - Determine the number of connected subnetworks in this network.
- Now, remove the link E-F and D-F from the network. How many connected subnetworks exist after these removals?
- If all connections become one-way (directed) in the order: $A \rightarrow B \rightarrow D \rightarrow F$, $A \rightarrow C \rightarrow E \rightarrow F$, $B \rightarrow C$, $D \rightarrow E$, can DFS starting from A reach all hubs? Provide a brief explanation.
 - Which change in the underlying data structure (queue $\leftrightarrow$ stack) would make BFS behave like

DFS?

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5B) Develop a C program using a Doubly linked list to manage a queue system in a hospital, where patients can register, be served in order, or leave the queue. (3)
- ```
struct Patient {
 int id;
 char name [50];
 struct Patient *next; };
```
- Function prototypes:  
void registerPatient(int id, char name[])  
void servePatient() //as per queue order  
void removePatient(int id)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 5C) Write a C function void rearrangeQueue(int queue[], int \*front, int \*rear) that rearranges the queue such that all even numbers appear before all odd numbers while preserving their original relative order within their respective groups (evens and odds). (2)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

-----End-----

| Event Name | Subject |
|------------|---------|
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058-END SEMSTER-ODD-2025 CSS\_2101:Data Structures

| #     | Course Outcome | Description | Marks |
|-------|----------------|-------------|-------|
| 1     | 1              | CO 1        | 9     |
| 2     | 2              | CO 2        | 8     |
| 3     | 3              | CO 3        | 12    |
| 4     | 4              | CO 4        | 16    |
| 5     | 5              | CO 5        | 5     |
| Total |                |             | 50    |

\* **NA** - CO Not Selected

| Event Name | Subject |
|------------|---------|
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| #     | Level       | No of Questions | Marks (%) |
|-------|-------------|-----------------|-----------|
| 1     | Remember    | 0               | 0(0%)     |
| 2     | Understand  | 0               | 0(0%)     |
| 3     | Application | 9               | 32(64%)   |
| 4     | Analysis    | 6               | 18(36%)   |
| 5     | Create      | 0               | 0(0%)     |
| 6     | Evaluate    | 0               | 0(0%)     |
| Total |             | 15              | 50 (100%) |